

Level 3 Calculus

91579 Apply integration methods in solving problems

Practice Problems (Merit Level)

Achievement	Achievement with Merit	Achievement with Excellence
Apply integration methods in solving problems.	Apply integration methods, using relational thinking, in solving problems.	Apply integration methods, using extended abstract thinking, in solving problems.

You may need the Formulae and Tables Booklet L3–CALCF.

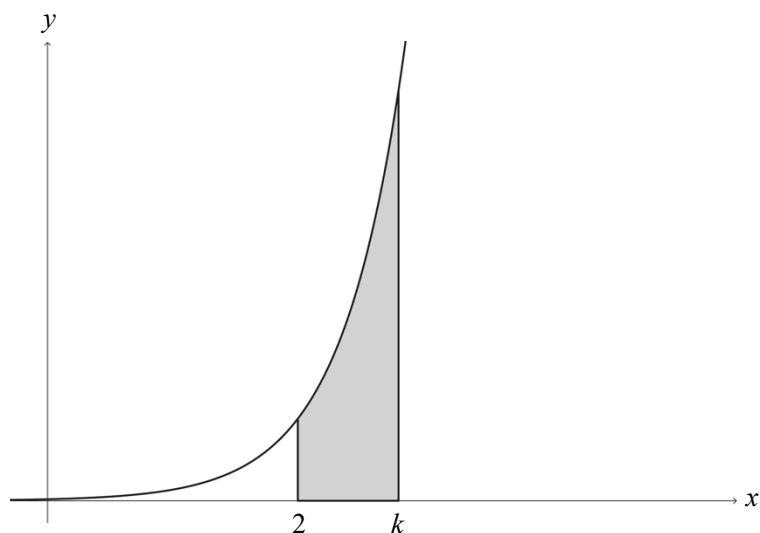
Show ALL working.

You must use calculus and show the results of any integration needed to solve the problems.

QUESTION ONE: DEFINITE INTEGRALS INVOLVING UNKNOWN CONSTANTS

(a) Find the value of p (where $p > 0$), given that $\int_{-6}^p \frac{3x+10}{x+7} dx = 3p$.

(b) The graph below shows part of the graph of the function $y = 5e^{2x-4}$.



Find the exact value of k such that the shaded region has an area of 10.

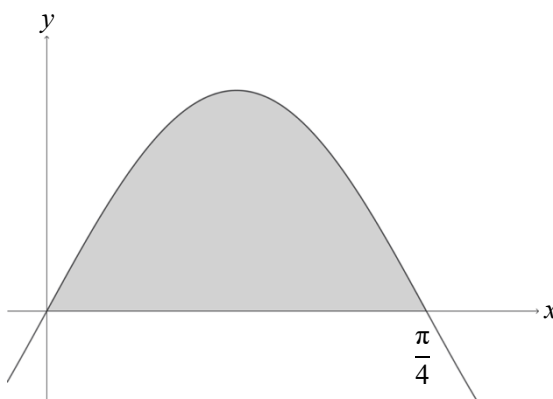
(c) It is given that $\int_0^m (3x^2 - 2nx) dx = -8$ and $\int_1^n \left(\frac{m}{x^2}\right) dx = m - \frac{1}{2}$, where m and n are constants and $m \neq 0$.

Determine the values of m and n .

(d) Find the value of k (where $k > 0$), given that $\int_0^k \frac{6x}{x^2 + 4} dx = \ln(125)$.

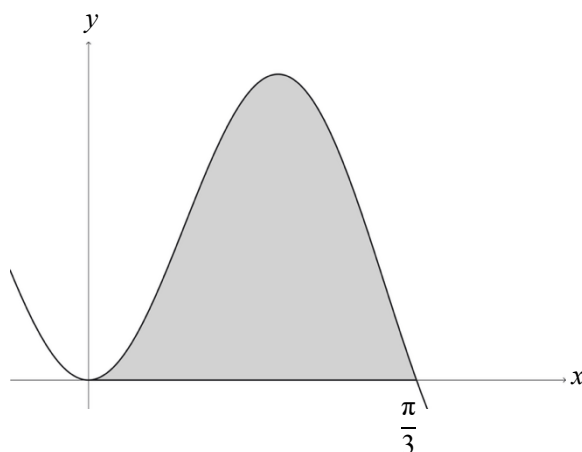
QUESTION TWO: AREA UNDER A CURVE & TRIGONOMETRIC IDENTITIES

(a) The graph below shows part of the graph of the function $y = 6 \sin(2x) \cos(2x)$.



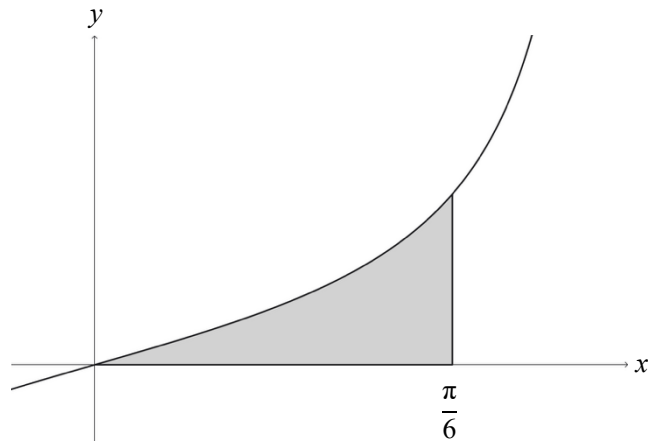
Find the shaded area under the curve between $x = 0$ and $x = \frac{\pi}{4}$.

- (b) The graph below shows part of the graph of the function $y = A \sin(3x) \sin(2x)$, where A is a positive constant.

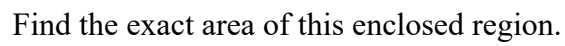


Find the value of A , given that the area of the region bounded by the curve and the x -axis between $x = 0$ and $x = \frac{\pi}{3}$ is $6\sqrt{3}$.

(c) The graph below shows part of the graph of the function $y = \frac{2 \tan x}{1 - \tan^2 x}$.



Find the shaded area under the curve between $x = 0$ and $x = \frac{\pi}{6}$.



QUESTION THREE: KINEMATICS

- (a) An object's acceleration can be modelled by the equation $a(t) = 1.5e^{0.5t}$, where a is the acceleration of the object in m s^{-2} , and t is the time in seconds from the start of timing.

Initially, the object has a velocity of 2 m s^{-1} .

How far did the object travel during the 3rd second of its motion?

- (b) An object's motion can be modelled by the equation $a(t) = 3 \cos(2t)$, where $t \geq 0$. a is the acceleration of the object in m s^{-2} , and t is time in seconds.

At $t = 0$, the object has a velocity of 1 m s^{-1} and a displacement of 2 m from a fixed origin point O .

What is the displacement of the object at time $t = \pi$?

- (c) An object's acceleration can be modelled by the equation $a(t) = \frac{6e^{3t}}{(e^{3t}+2)^2}$, where a is the acceleration of the object in m s^{-2} , and t is the time in seconds from the start of timing.

At $t = 0$, the object has a velocity of 2 m s^{-1} .

Find the exact velocity of the object when $t = \ln(2)$ seconds.

QUESTION FOUR: DIFFERENTIAL EQUATIONS

- (a) Consider the differential equation $\frac{dy}{dx} = \frac{4e^{2x}}{y^2}$.

Given that $y = 3$ when $x = 0$, find the value of y when $x = \ln(2)$.

(b) Consider the differential equation $\frac{dy}{dx} = y^2 \sin(3x)$.

Given that $y = 1$ when $x = \frac{\pi}{3}$, find the value of y when $x = \frac{\pi}{6}$.

(c) Consider the differential equation $\frac{dy}{dx} = 16 \sin(3x) \cos(5x)$.

Given that $y = 5$ when $x = \frac{\pi}{2}$, find the value of y when $x = \frac{\pi}{4}$.

(d) Consider the differential equation $\frac{dy}{dx} = \frac{x}{y\sqrt{x^2+16}}$, where $y > 0$.

Given that $y = 4$ when $x = 3$, find the value of y when $x = 0$.

QUESTION FIVE

(a) Show that the exact value of the definite integral below is $\frac{\sqrt{2}}{4}$.

$$\int_0^{\frac{\pi}{8}} \frac{1 - \tan^2(x)}{\sec^2(x)} dx$$

(b) Use the substitution $u = e^x + 1$ to find the exact value of the definite integral:

$$\int_0^{\ln(3)} \frac{e^{2x}}{e^x + 1} dx$$
